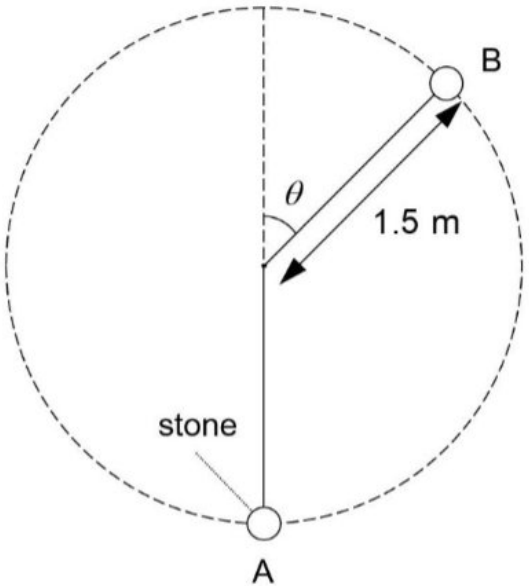
7 A stone attached to a light inextensible string at position A is given an initial push to the right. It subsequently moves in a vertical circle of radius 1.5 m and the string just slackens at position B when it makes an angle of θ to the vertical.



Given that the velocity of the stone at position B is 2.2 m s^{-1}, what is θ ?

A 9° **B** 19° **C** 71° **D** 81°